Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/02/2005
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Project Title:	Confocal microscopy of the colloidal glass transition		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
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Task Description:	We study concentrated colloidal suspensions, a model system which has a glass transition. By using a confocal microscope to follow the three-dimensional motion of colloidal particles, we can directly observe the microscopic behavior responsible for the macroscopic viscosity divergence of glasses. To use direct imaging to quantify the microscopic behavior of particles as the colloidal glass transition is approached. Specifically, both structural properties and dynamical properties are to be measured and compared with theoretical descriptions of the glass transition. We also seek to characterize the aging of colloidal glasses; we wish to determine how colloidal glasses change over time. The overarching goal is to clarify the mechanisms underlying the glass transition and to provide key data to test various theories of the glass transition. Confocal microscopy of the colloidal suspensions allows us to both directly follow the motion of individual particles and to visualize structural arrangements as the transition is approached. These		

	measurements are crucial to testing and extending theoretical understanding of both dense colloidal suspensions and glasses in general. Moreover, the insight from these experiments will be useful to the processing of industrially important materials. Finally, by understanding how the glass transition occurs in these model systems and how the dynamics are affected by the system details, we may be able to design novel materials with specific properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this year of funding, we finished up our experiments on perturbing dense colloidal suspensions with magnetic beads. Most recently, we've focused on writing up our experimental results. This mostly involved work by the PI, and Prof. Piotr Habdas (St. Joseph's University) who was previously funded as a postdoctoral researcher by this grant. Additionally, the experimental work was continued this quarter by an undergraduate, Ben Bluth, and related data analysis was conducted by a graduate student, Gianguido Cianci. Dense colloidal suspensions are a simple model system, which becomes glassy when the colloidal particle concentration is above a critical threshold. Below this threshold, the sample is analogous to a liquid, and colloidal particles move around the sample due to Brownian motion. The glass transition is identified at the point where the diffusion of colloidal particles through the sample can no longer be observed on reasonable experimental time scales. The goal of our experiments is to observe how such samples respond to being locally perturbed in a variety of ways. In particular, we wish to characterize how the response changes in liquid-like samples as the transition is approached. We have investigated how the sample responds when it is stirred by a tiny magnetic stir bar. We place a very small concentration of 5 micron diameter magnetic particles into the sample of 2 micron diameter PMMA colloidal particles. Occasionally, the magnetic particles stick together, forming dimers. We put a permanent magnet on a stepper motor above the sample, and by rotating the magnet can rotate the dimer and stir the sample locally. Our preliminary data indicate that there is a stirring region of finite radius. Dr. Habdas is analyzing this data to determine how the radius of the perturbed region changes as the colloidal concentration is increased; we have hints that this radius grows. Gianguido Cianci has been analyzing this data to determine how regions of the sample are stretched in 3D by the stir bar. His analysis has discovered that the experiment takes a long time (several dimer rotations ? hours) to reach steady-state, and that the static properties of the sample evolve over time until this steady-state is reached. Doug Anderson studied what happens when the force applied to an isolated single magnetic bead is removed and reapplied. The force duration is short enough that permanent rearrangements do not occur; however, the surrounding colloidal particles do move slightly and then return to their original locations. Each time, we find the particles move exactly as if they are in a homogeneous, isotropic elastic medium. This is surprising to us, as the magnetic bead is of comparable size to the surrounding colloidal particles, and thus on these length scales the medium is definitely not homogeneous – we expected that the discrete individual particles would be perturbed heterogeneously in space. Mr. Anderson also took data showing how the sample relaxes when the force has been removed; our data show that this depends sensitively on the initial displacement of the probe. Rachel Courtland studied the behavior of a magnetic bead that is slowly oscillated in a circle. Because the forcing frequency is known, Fourier techniques are proving useful in the data analysis. An additional advantage of this technique is that the sample is not permanently disturbed. The data can be analyzed according to the ideas of "microrheology" to determine the bulk viscoelastic properties of the sample, and her results are in reasonable agreement with previous observations. Furthermore, the response of the surrounding colloidal particles again indicates that the sample is acting like a homogeneous isotropic medium, in agreement with Mr. Anderson's data. Her work resulted in her master's thesis, which she successfully defended in August 2004. Ben Bluth, an undergraduate, and Dr. Vikram Prasad, a postdoctoral researcher, are examining what happens when a magnetic bead is pulled with a large constant force. We published data on this in 2004 for a particular colloidal suspension where the magnetic bead was twice the diameter of the colloidal particles. Dr. Prasad's experiments examine the force/velocity relationship when we vary the size ratio between the magnetic bead and the colloidal particles. Mr. Bluth's experiments are studying what happens in a binary mixture of two particle sizes, where crystallization will not be a problem. Thus, these samples will allow us to conduct very long duration experiments, previously prevented by crystallization. In particular we'll be able to study magnetic beads moving at extremely slow velocities. Their preliminary data indicate that we get similar results to our earlier work. Gianguido Cianci (a graduate student) is studying the aging of colloidal glasses. By stirring a dense sample vigorously, the sample is initialized, and the subsequent dynamics show aging: the time scale for particle motion grows with the age of the sample (the time since stirring ended). In our first year of funding, we studied the aging in monodisperse colloidal samples (resulting in one publication so far). Currently, Mr. Cianci is looking for structural changes that take place in the sample upon aging, similar to what he saw in the dimer stirring data from Piotr Habdas.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Habdas P, Schaar D, Levitt AC, Weeks ER. "Forced motion of a probe particle near the colloidal glass transition." Europhys. Lett. 2004 Aug;67(3):477-83. http://dx.doi.org/10.1209/epl/i2004-10075-y , Aug-2004
Awards	Weeks, E.R. "Presidential Early Career Award for Scientists and Engineers, May 2004." May-2004
Dissertations and Theses	Courtland, R.E. "Oscillatory perturbations of dense colloidal suspensions." Master's Thesis, Emory University, August 2004. , Aug-2004
Presentation	Weeks, E.R. "Poking colloidal glasses" Granular / Granular-Fluid Flow Gordon Research Conference, Colby College, ME Jun-2004
Presentation	Weeks, E.R.; Courtland, R.E.; Habdas, P. "Oscillatory local perturbations of dense colloidal suspensions" AIChE Annual Meeting, Austin, TX Nov-2004

Presentation	Weeks, E.R. "Local perturbations of colloidal glasses" 2nd Workshop on Complex Systems, Sendai, Japan Dec-2004
Significant Media Coverage	Covington, M. "The feel of food fascinates Emory researcher." Interview on WABE FM 90.1 (NPR) in Atlanta, January 28, 2004., Jan-2004